

Generative AI, kept behind a verifier

Step 9, the bonus. Turning a number a tutor cannot read into a sentence they can act on.

Jeryl Donato Estopace

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The explanation layer for the dropout model, and the check that makes it safe

A tutor cannot read a SHAP value

This is everything the model knows about one flagged student, and it has to become two plain sentences.

Tuition fees up to date	+2.50
Course	+1.98
mother's occupation	+1.54
Admission grade	+0.71
mature entry	+0.68

The top drivers behind one student's risk score, straight from Step 5.

A language model can turn this into a note.

The risk is obvious. Handed the numbers, it will print a score and name things about a person that Step 5 spent a whole notebook forbidding, so the language model never runs alone and always behind a check.

Ask it with no rules, and it does harm

This is the naive prompt on that same student, with the forbidden things marked.

It writes that the admission grade is **95.1**, that the student is in **Course 9119**, and that the mother's occupation code of **0** is a proxy for **lower socioeconomic status**, and it concludes they are **likely** to struggle.

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notes safe to read to a student

That is a string of numbers, a course code, and an unearned claim about the student's mother. On another student the model even named their gender outright, and every one of these is a thing Step 5 forbids.

A prompt is a request. The verifier is the control.

Same student, same data. Four rules, and a cell that checks they held.

This student is facing some pressure due to outstanding tuition fees and may need extra support navigating their course. A warm check-in to see how they are settling in.

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notes safe to read to a student

The four rules are simple. The model uses no numbers and no scoring words, names no driver that is not in this student's own list, and never mentions gender even though it leans on it. The rules-engine baseline needs none of them, because it can only say what the table already holds.

Point it at any student

explain_student(582). The rules-engine note and the language-model note, side by side.

student 582 (actually dropped out)

rules-engine note:

Behind on tuition fees, and came in through the mature-entry route. Ask whether the payment schedule is workable, and check they know the hardship fund exists.

guarded language-model note:

Facing some pressure from outstanding tuition fees, and may need extra support

A tutor names a student, and gets a sentence they can act on. It runs off a committed cache, so a grader needs no key.

What it bought, and what it did not

The context is the work. The generation is the easy part.

It bought fluency, and words for the course code and the parent's occupation the table refuses to touch.

It also bought one note that over-reached, turning a parent's occupation into a claim about their schooling. The table declined to make it, and the verifier is not built to catch it. So the verifier is necessary, not sufficient.

Everything the model knows about which number matters, it knows because it was told, in a context block that took the rest of the project to earn.

Generative AI did not replace the fairness audit. It sits on top of it, and it is only trustworthy because that came first.